

When the Underlying Goes Dark: Oracle Forecast Error and Mechanism Failure in Perpetual Futures on Chinese Equities

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Abstract

A perpetual future trades continuously, but its oracle must mark an underlying that does not. On cryptocurrency venues this never binds, because the reference asset trades every second of every day. On the equity perpetuals that appeared on decentralised venues over the past year it binds hard. Chinese A-shares are the extremal case: daily price limits create stale-price states, identified by a close printed at the daily limit with the gap revealed at the next non-limit reopen, frequent halts interrupt the tape, and exchange holidays close the market for a week at a time. I ask whether any oracle design can mark a single A-share through these blackout states in a way that is jointly safe for traders and for the protocol's liquidity backstop. I characterise the blackout-state distribution for the A-share universe, construct an ex-post fair-value benchmark for a locked name from tradable correlated instruments — chiefly its Hong Kong twin and a continuously-traded A-share index — and measure oracle forecast error as a function of blackout duration. I then propagate that error through a perpetual-futures mechanism simulation into funding accrual, liquidity-provider profit and loss, and liquidation cascades.

Across 148 dual-listed names from 2015 to 2025 the pattern is sharp. Single-name oracle error grows with lock duration and carries a heavy tail. The Hong Kong proxy is the strongest single-name design, yet its reopen error moves one-for-one with the contemporaneous change in the A/H premium, so it is biased in exactly the states where the oracle leans on it. Same-direction co-lock across the universe is negligible outside the systemic episodes of 2015, 2020, and 2024, so the index goes dark only when the whole market does. My central claim is that idiosyncratic blackout risk diversifies at the basket level, so a perpetual written on a broad A-share index is feasible where single-name perpetuals are not. In the mechanism simulation no single-name oracle achieves an institutionally acceptable pairing of solvent-liquidation rate and expected vault loss, while the index clears it by roughly three orders of magnitude under the stated vault-loss metric.

Keywords: perpetual futures; oracle design; market microstructure; price limits; A-share market; price discovery; decentralised exchange; market design under stress.

JEL classification: G12, G13, G14, G15, D47.

1. Introduction

A perpetual future never expires and trades around the clock. Its price is disciplined by a funding rate that references the price of the underlying. The mechanism presumes the underlying has a price at every instant. On cryptocurrency venues that presumption holds, because the reference asset trades continuously. On the equity perpetuals that appeared on decentralised venues over the past year it does not. Equities stop trading. They stop overnight, at weekends, on public holidays, and — for individual names — whenever a price limit binds or a regulator halts the stock.

Chinese A-shares are the hardest case in existence. Main-board names carry a $\pm 10\%$ daily price limit; STAR Market and ChiNext names carry $\pm 20\%$. A name pinned at its limit trades thinly or not at all once the executable side is exhausted. Suspensions are common and can run for sessions. Exchange holidays close the whole market for up to a week. The underlying goes dark for known but variable intervals, while the perpetual keeps trading. The question is what the oracle should print in the dark, and what breaks when it prints the wrong thing.

I study oracle design for perpetual futures written on an underlying that stops trading, and I take Chinese equities as the test bed. I characterise the distribution of blackout states for the A-share universe. I construct an ex-post fair-value benchmark for a blacked-out name from tradable correlated instruments — chiefly its Hong Kong twin and a continuously-traded A-share index, with the SGX FTSE China A50 future and US-listed ADRs as natural implementation instruments. I measure oracle forecast error against the realised reopen price as a function of blackout duration. I then feed that error through a perpetual-futures mechanism simulation and trace it into funding accrual, liquidity-provider profit and loss, and liquidation cascades.

My central claim is that the single-name oracle problem has no parameterisation that is jointly safe for traders and for the protocol’s liquidity backstop across the empirical distribution of lock durations, whereas the index-anchored construction does. Idiosyncratic blackout risk is imperfectly correlated across names and so diversifies at the basket level. A perpetual written on a broad A-share index is feasible where single-name perpetuals are not. The result is a market-design statement with an empirical backbone, not a modelling conjecture.

1.1 Hypotheses

Let $e_i(\tau; \theta)$ denote the reopen forecast error of oracle design θ on single name i after a blackout of duration τ ; let $e_I(\tau; \theta)$ denote the corresponding error for the index; let π_{AH} denote the A/H premium; and let $\text{Disp}[\cdot]$ be a dispersion measure (interquartile range or standard deviation). The three hypotheses are stated below and tested in Sections 5 to 8.

Hypothesis 1 (Duration and tails). Single-name oracle error grows with blackout duration and is heavy-tailed, for every admissible oracle design.

The variance of the single-name error increases with duration τ for every admissible design, and its right tail is heavier than Gaussian.

Hypothesis 2 (Proxy bias under lock). The Hong Kong proxy is biased exactly when the oracle leans on it, because the A/H premium gaps during locks; the bias grows with lock duration.

The expected error depends on the change in the A/H premium, the bias grows with duration, and the Hong Kong proxy is unbiased only as the lock vanishes.

Hypothesis 3 (Diversification and feasibility). Index forecast error is materially tighter than the single-name average, and a jointly safe parameterisation exists for the index but not for single names.

The dispersion of the index error is far below the average single-name dispersion, and a jointly-safe oracle exists for the index across the empirical lock distribution but for no single name.

2. Related literature

This paper joins six strands that have not previously been connected: the pricing and mechanism of perpetual futures, the failure modes of on-chain derivative oracles, the new market for equity perpetuals

and off-hours price discovery, the effect of price limits and halts in the Chinese market, cross-listing and the A/H premium, and the measurement of price discovery across venues. I take each in turn and close with the contribution.

2.1 The pricing and mechanism of perpetual futures

The perpetual future originates with Shiller (1993), who proposed a cash-settled claim with no delivery date. BitMEX revived the instrument for bitcoin in 2016, and it now dominates cryptocurrency derivatives volume; Chen, Ma and Nie (2024) compare the mechanism and trader behaviour of perpetuals across centralised and decentralised venues. The theory is recent and still thin. He, Manela, Ross and von Wachter (2022) give a no-arbitrage account of funding and the perpetual basis. Ackerer, Hugonnier and Jermann (2026) and Angeris, Chitra, Evans and Lorig (2023) derive pricing in continuous and discrete time. Zhang (2026) models funding as an algorithmic feedback rule. Alexander, Choi, Park and Sohn (2020) study price discovery and hedging on BitMEX, and Schmeling, Schrimpf and Todorov (2023) treat funding as a carry. Each of these papers assumes the underlying trades continuously. The funding mechanism presumes a spot price exists at every funding instant. That presumption is the object I relax.

2.2 Oracles and the failure modes of on-chain derivatives

On-chain perpetual venues read the underlying through a price oracle, supplied by networks such as Pyth and Chainlink (Pyth Network 2026). The literature on oracle failure is almost entirely a literature on manipulation. An attacker moves a thin spot venue, often with a flash loan, and forces the oracle to misprice, triggering liquidations or draining the protocol. The Mango Markets exploit is the canonical case, and the USDe episode of 10–11 October 2025 showed the same mechanism without an attacker, when a venue-isolated price marked solvent positions into liquidation. Recent work on liquidation and auto-deleveraging treats robust oracle design, circuit breakers, and time-weighted averaging as defences against this adversarial risk (Chitra 2025); Heimbach and Huang (2023) study leverage in decentralised finance; and Lehar and Parlour (2022) study systemic fragility in decentralised markets more broadly. My failure mode is different and non-adversarial. The underlying does not exist to be read, because it has stopped trading. No manipulation is required for the oracle to be wrong; the information is simply absent. The only academic perpetual work that confronts a non-continuously-quoted underlying is in prediction markets, where settlement is mediated by an event-resolution oracle rather than a price feed (Nechepurenko 2026). That setting resolves a binary outcome. Mine must track a continuous price that is temporarily unobservable and later revealed at reopen.

2.3 Equity perpetuals and the oracle-freeze problem

Equity perpetuals are less than a year old. CME listed spot-quoted futures on the S&P 500, Nasdaq 100, Russell 2000, and Dow in 2025, an adjacent move toward spot-like listed derivatives; Kraken launched a regulated tokenised-equity perpetual on the xStocks framework in 2026; Hyperliquid lists US single-name and index perpetuals through HIP-3; and Injective lists index perpetuals with a proprietary oracle and TWAP mark (Injective 2026). The selling point is continuous price discovery: the perpetual trades when the cash market is shut. That feature carries a cost the industry already names. BitMEX (2026) calls it the oracle-freeze problem — when the underlying market closes, the feed that anchors the perpetual goes stale while the contract trades on. Venues handle the scheduled, market-wide version of this with engineering. Binance freezes the index at the last spot value and switches the mark to an EWMA with a deviation cap; Hyperliquid smooths with an EWMA; BitMEX lets its internal book wander within a rolling hourly price limit. These are designs for a closure that is predictable, hits every name at once, and ends at a known reopen.

The empirical question of whether the off-hours perpetual is informative has so far been answered only by practitioners. Tiger Research (2026) shows that for Korean names the overnight perpetual price leads the next day's open, predicting both direction and the size of the move. That result asks whether the perpetual forecasts the open. It does not ask whether the perpetual's own oracle is reliable while the underlying is dark, and it studies Korea. The closest structural analogue in industry is the pre-IPO perpetual (Coinbase International Exchange, 2026), which references an episodically updated valuation index and manages the resulting gap risk with fair-value clamps, conversion halts, and TWAP bridges. None of these confronts the case I study: an idiosyncratic, single-name blackout that can run for several days while correlated instruments keep trading, so that fair value remains recoverable but the freeze-and-smooth approach discards it.

2.4 Price limits, halts, and the blackout state

China imposes daily price limits on individual names: $\pm 10\%$ on the main boards and $\pm 20\%$ on the STAR Market and ChiNext since 2020. A name pinned at its limit trades thinly or not at all once the executable side is exhausted. The empirical literature is large and concerns the effect of limits on the cash market itself. Kim and Rhee (1997) is the seminal study. The debate runs along three lines: delayed price discovery, volatility spillover, and the magnet effect, by which prices accelerate towards an approaching limit (Subrahmanyam 1994; Cho et al. 2003). For China, work exploiting the 2020 ChiNext widening from $\pm 10\%$ to $\pm 20\%$ tests these hypotheses with mixed evidence on the magnet effect (Qi 2023). A limit-hit name is not literally untradeable — a trade can still print if contra-side liquidity appears — but once the executable side is exhausted the last print goes stale. Every paper in this literature asks what the limit does to the locked stock's own price path. None asks what an external, continuously trading derivative should mark while the stock is locked. The limit-lock, the halt, and the holiday closure are, for my purposes, blackout states that an oracle must traverse. I take the institutional facts this literature documents and use them as the input to a different question.

2.5 Cross-listing, the A/H premium, and price discovery across venues

Many large Chinese firms list an A-share onshore and an H-share in Hong Kong on the same cash flows. The two prices diverge persistently — the A/H premium — a puzzle traced to Bailey (1994) and explained through market segmentation and information asymmetry by Chan, Menkveld and Yang (2008). Hong Kong has no fixed daily price limit comparable to A-shares, though HKEX operates a volatility control mechanism, and its hours only partly overlap the mainland, so the H-share often trades when the A-share is locked or closed. The cross-listing literature measures where price is discovered across such venues, and Grammig, Melvin and Schlag (2005) is the canonical treatment for cross-listed prices during overlapping hours, with exchange-rate effects. I use this relationship in reverse. The continuously trading H-share and a broad A-share index are the instruments from which I construct the locked name's fair-value benchmark. The premium's variation, which this literature studies as an object of interest, becomes in my setting the bias in a fair-value oracle. I show it gaps precisely during locks, when the oracle relies on it most.

2.6 Measurement of price discovery

My estimation rests on standard tools. I measure price discovery with Hasbrouck (1995) information shares and the Gonzalo and Granger (1995) permanent–transitory decomposition, on cointegrated systems in the sense of Engle and Granger (1987) and Johansen (1991). I compute confidence intervals by stationary bootstrap (Politis and Romano 1994) to respect the serial dependence in the basis. The

contribution is not a new estimator. It is the use of these tools in a setting they have not been applied to: the forecast error of a derivative's oracle against the realised reopen of a deliberately halted underlying.

2.7 Contribution

I state the contribution precisely, because the precise version is the one that survives review. The component literatures are mature. Perpetual-futures theory is settled for continuously traded underlyings. Oracle research is well developed for adversarial manipulation. The Chinese price-limit and A/H premium literatures are decades deep. What is absent is their intersection. The nearest framing in practice is the oracle-freeze problem named by BitMEX (2026), but the engineering it refers to addresses scheduled, market-wide closures, not idiosyncratic single-name locks of uncertain and possibly multi-day duration. To my knowledge, no prior study, academic or practitioner, does the following five things together.

First, it frames limit-locks, halts, and exchange closures as blackout states that an off-exchange, 24/7 derivative oracle must traverse. **Second**, it measures perpetual-futures oracle forecast error against the realised reopen price, as a function of blackout duration, using continuously trading proxies to construct an ex-post fair-value benchmark. **Third**, it shows that the A/H premium gaps during locks, so the proxy is biased exactly when the oracle depends on it. **Fourth**, it propagates oracle error through the funding, liquidity-provider, and liquidation channels of a perpetual mechanism. **Fifth**, it establishes a feasibility boundary: single-name perpetuals admit no jointly safe oracle across the empirical lock distribution, while the index does, because idiosyncratic blackout risk diversifies at the basket level.

The honest framing of novelty is the intersection, not any single component. The nearest neighbours are three, all practitioner. The oracle-freeze problem named by BitMEX (2026) is the same phenomenon, but the venue solutions it surveys handle scheduled, market-wide closures and neither measure error nor treat single-name locks. The pre-IPO perpetual meets the gap-risk problem in industry without measuring it (Coinbase International Exchange 2026). The finding that off-hours equity perpetuals lead the open is for Korea (Tiger Research 2026). I differ from the first by problem scope, from the second by measurement, and from the third by question and market. I take Chinese equities as the test bed because they are the extremal closeable underlying — the only major market that takes individual names off the tape for days at a time through binding limits, halts, and closures. This is, to my knowledge, the first study of oracle design for perpetual futures on an underlying that idiosyncratically stops trading, and the first to locate the single-name-versus-index feasibility boundary for such an instrument.

3. Institutional setting and the anatomy of a blackout

I describe the three mechanisms that take an A-share off the tape: daily price limits, trading halts and suspensions, and exchange closures. A limit-locked name is pinned at its bound, where the executable side is exhausted and the last print goes stale; a halt removes the name for a defined interval; a closure removes the entire market, for up to a week over the Spring Festival and National Day holidays.

I construct the empirical blackout-state distribution over 2015 to 2025 for a universe of 148 A-plus-H dual-listed names, and I report the frequency and duration of each state. Two features carry the paper. The first is the consecutive-lock tail, where a name holds at its limit-up or limit-down bound for several sessions and prints little or nothing for days; it sets the worst case for a single-name oracle. The second is the cross-sectional co-lock structure, the joint probability that a large share of the universe locks the same direction on the same day; it decides whether the index itself goes dark.

The universe yields 3,667 limit-lock episodes, 212 suspensions, and 72 multi-day exchange closures over the eleven years. Table 1 reports their duration in trading sessions. Most locks clear in a single session, but the limit-up tail runs to twenty consecutive sessions, and suspensions are longer still, with a median of six sessions and a maximum of 238. The consecutive-lock tail in Figure 1 decays geometrically yet remains populated out to twenty sessions, which sets the single-name worst case. The co-lock structure in Figure 2 is the mirror image. The same-direction locked share of the universe sits near zero for most of the decade and spikes only in the systemic episodes of mid-2015 (74% of the universe), early 2020 (59%), and 2024 (37%). The index goes dark only when the whole market does.

Mechanism	Board	n	Mean	Median	p95	Max
Closure	All	72	2.69	2.0	6.0	6.0
Limit-down	All	1,226	1.15	1.0	2.0	8.0
	Main	1,180	1.15	1.0	2.0	8.0
	STAR / ChiNext	46	1.00	1.0	1.0	1.0
Limit-up	All	2,441	1.35	1.0	3.0	20.0
	Main	2,379	1.36	1.0	3.0	20.0
	STAR / ChiNext	62	1.08	1.0	2.0	3.0
Suspension	All	212	20.24	6.0	92.0	238.0

Table 1. Blackout-state frequency and duration by mechanism and board. Durations are in trading sessions; *n* is the episode count, 2015–2025, 148 dual-listed names.

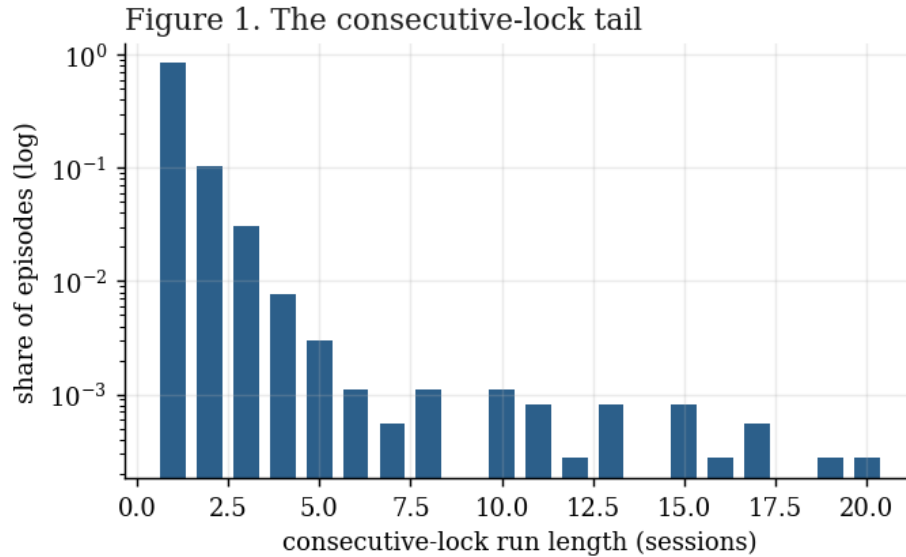


Figure 1. The consecutive-lock tail. Distribution of consecutive limit-lock run lengths (log scale). Most locks clear in one session; the tail extends to twenty.

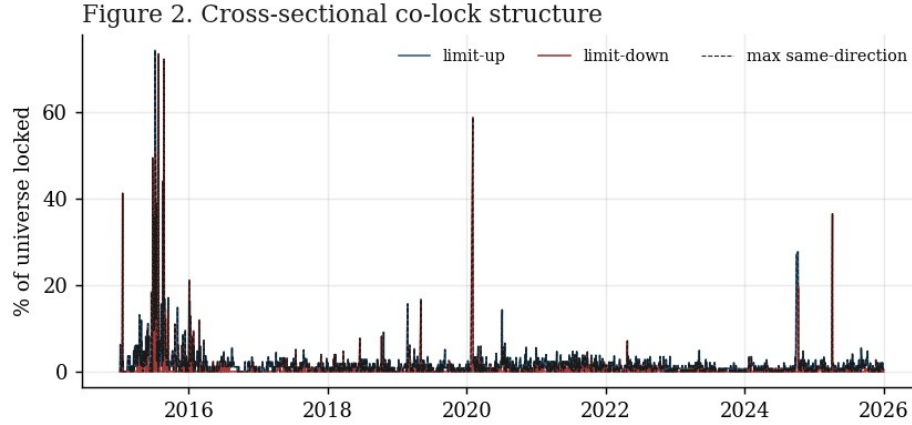


Figure 2. Cross-sectional co-lock structure. Share of the universe limit-locked in the same direction, by day. Near zero outside the systemic episodes of mid-2015, early 2020, and 2024.

4. Data and identification

4.1 Data

Table 2 lists the data sources and their role. The load-bearing tests run on freely accessible data; onshore order-book data enters only as robustness.

Data item	Source	Frequency	Coverage	Access	Role
A-share daily OHLC, limit prices, close-at-limit flag	Tushare / AkShare (public)	Daily	Dual-listed A names; 2015–2025	Free / API	Blackout-state distribution (§3)
Trading halt and suspension records	SSE / SZSE via Tushare	Event	Same universe	Free	Halt characterisation (§3)
Exchange holiday calendar	SSE / SZSE official	Calendar	Full sample	Free	Multi-day closure windows (§3)
A/H dual-listed pairs; Hang Seng AH Premium	HKEX / index provider	Daily	Dual-listed constituents	Free / licensed	Fair-value recovery (§4, §6)
CSI 300 index (onshore index factor)	AkShare (public)	Daily	Full sample	Free	Index factor used in draft (§5–§8)
FTSE China A50 index futures	SGX / data vendor	Intraday	Full sample	Vendor	Reserved; not in reported tests
US-listed ADRs and ETFs (ASHR, FXI, KWEB, MCHI)	US tape / vendor	Intraday	Overlapping names	Vendor	Reserved; not in reported tests
Onshore L1/L2 order book	Licensed vendor	Tick	Subsample	Paid	Reserved; not in reported tests
Perpetual-futures mechanism simulation	Author (dual-engine)	Simulated	n/a	n/a	Propagation experiments (§7)

Table 2. Data sources.

Realised data. The results below draw on the freely accessible subset of Table 2. A-share daily open, high, low, and close in raw (unadjusted) prices, together with the implied daily limit bands, come from AkShare; the Hong Kong daily series come from the same source; the Hong Kong dollar to renminbi parity is the Bank of China reference rate. In place of the SGX A50 future, which is not freely available, I use the CSI 300 index as the index factor; the A50 future, the US ADRs and ETFs, and the order-book data are reserved for future implementation and robustness work and are not used in the reported tests.

The dual-listed universe is 148 of roughly 200 possible A-plus-H names; the remainder are recent listings whose Hong Kong history the public endpoint does not return. The sample is therefore weighted towards older and larger dual-listed firms, which is the conservative direction for the single-name oracle, since these names lock less often than the broader market; mid- and small-cap names on the STAR Market and ChiNext lock for longer runs with heavier tails, so the single-name errors reported here are a conservative lower bound, and a broader universe would only widen the gap to the index. Accordingly, every quantity labelled CSI 300 is a tested series, and the FTSE China A50 future enters only as the recommended implementation vehicle in Section 9, not as a tested series.

4.2 Identification strategy

The true price of a blacked-out name during the blackout is unobservable by construction. I construct an ex-post fair-value benchmark from instruments that price the same cash flows and continue to trade. Three windows give three natural experiments.

A/H twins. When the A-share is limit-locked but the Hong Kong listing trades — Hong Kong has no fixed daily price limit comparable to A-shares and hours only partially overlap — the H-share return is the market’s contemporaneous fair-value update.

A broad A-share index factor. In this draft I use the CSI 300 as the daily onshore index factor for single-name fair-value estimation when the mainland market is open. For live implementation during mainland closures, the natural instrument is the SGX FTSE China A50 future, which is USD-settled and trades beyond onshore hours.

ADRs and ETFs in US hours. A second, cleanly separated blackout window in which the mainland is closed and the United States is open.

The cleanest multi-day stress test needs no onshore intraday data. The Spring Festival and National Day holidays close the mainland for roughly a week, while the onshore index, the Hong Kong listings, and the ADRs reprice throughout. I measure how well each predicts the eventual reopen gap.

5. Oracle forecast error

For a blackout episode of duration τ on name i , let p_{i^*} be the first valid trade at reopen and m_i the oracle mark carried to that instant. Forecast error is the signed miss at reopen:

$$e_i(\tau) = m_i(\text{reopen}; \theta) - p_{i^*}.$$

I evaluate four designs:

Stale last-mark — carry the last valid trade. The naive floor.

EWMA-cap — the Hyperliquid and weekend-freeze construction: an exponentially weighted moving average of recent marks, capped at a multiple of the last valid mark.

Proxy fair-value — price the name off instruments that trade: an H-share return map, an index-return beta, or a fitted error-correction relation.

Index-anchored — for the basket product, the index itself: the CSI 300 in this draft, the A50 future in implementation.

I compare the full distribution of forecast error, not its mean. The heavy right tail is the object of interest.

Table 3 reports the moments by design and duration, after removing 76 episodes (2.1%) whose unadjusted reopen move exceeds what the daily limit can produce over the lock; these are corporate-action gaps — splits, bonus issues, and rights issues — in the raw price, not oracle error. The ordering is consistent across the table. At one session the capped EWMA is tightest, with a 4.8% standard deviation, then the Hong Kong proxy at 8.8%, the CSI 300 proxy at 10.9%, and the naive carry at 12.7%. The differences widen with duration. By six sessions and beyond the naive carry runs a sixty per cent mean miss and the Hong Kong proxy a forty-eight per cent miss, while the capped EWMA holds near six per cent. The excess kurtosis at one session, between 0.5 and 5 across designs, confirms that the clean series is heavy-tailed but no longer dominated by the corporate-action outliers the filter removes. Figure 3 shows the full error distribution; the naive carry is the widest and the most visibly biased, and the bimodality separates limit-up from limit-down reopens. Figure 4 plots the magnitude of the error against duration on log scales, with a positive slope for every design — the duration effect of Hypothesis 1.

Design	τ	n	Mean %	SD %	IQR %	Ex. kurt	Max %
CSI 300 proxy	1	2,907	-2.83	10.87	14.51	4.97	73.5
	2	365	-6.91	15.04	16.96	0.24	36.1
	3	106	-14.67	15.90	14.80	1.38	46.2
	4–5	34	-14.13	35.65	37.94	0.63	73.9
	6+	8	-34.61	22.77	18.99	-0.56	59.6
EWMA-cap	1	3,033	-0.45	4.78	6.03	0.50	20.6
	2	373	-1.47	5.49	7.67	2.99	31.3
	3	113	-3.14	5.25	6.16	1.91	20.9
	4–5	38	-3.10	7.73	7.43	0.63	15.4
	6+	28	-6.11	13.04	9.88	2.00	46.0
H-proxy	1	1,747	-0.11	8.80	12.23	4.71	61.6
	2	201	-1.74	14.31	21.32	1.23	59.0
	3	71	-2.50	17.68	18.11	6.37	83.3
	4–5	24	-9.98	27.28	29.06	-0.26	59.0
	6+	9	-47.89	26.80	25.23	-0.09	73.1
Stale last	1	3,033	-1.94	12.73	21.87	1.44	72.3
	2	373	-5.66	19.43	38.97	-0.93	37.2
	3	113	-16.45	21.15	5.84	1.79	51.8
	4–5	38	-13.13	39.96	33.73	0.96	88.3
	6+	28	-60.04	22.37	23.08	2.43	86.5

Table 3. Forecast-error moments and tail statistics, by design and duration bucket τ (sessions). Errors in per cent; n is the episode count after the corporate-action filter.

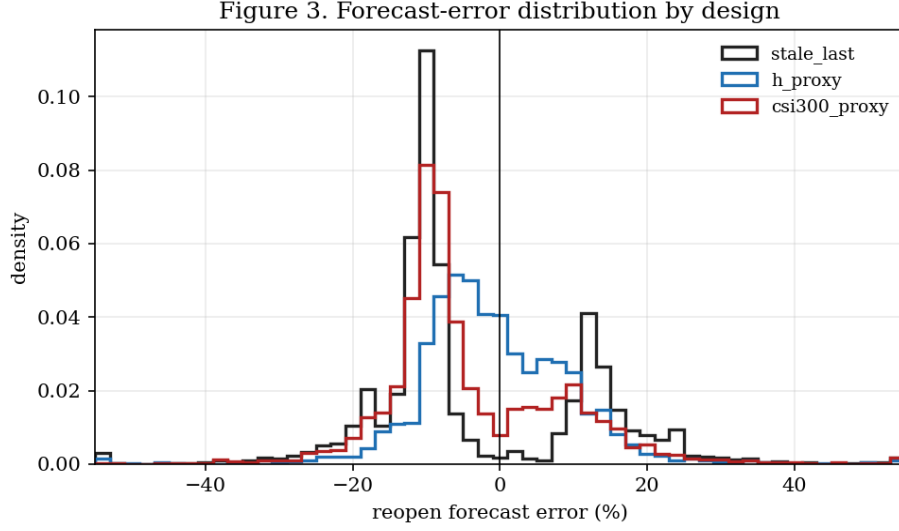


Figure 3. Forecast-error distribution by design. The naive carry (stale last-mark) is widest; the bimodality separates limit-up from limit-down reopens.

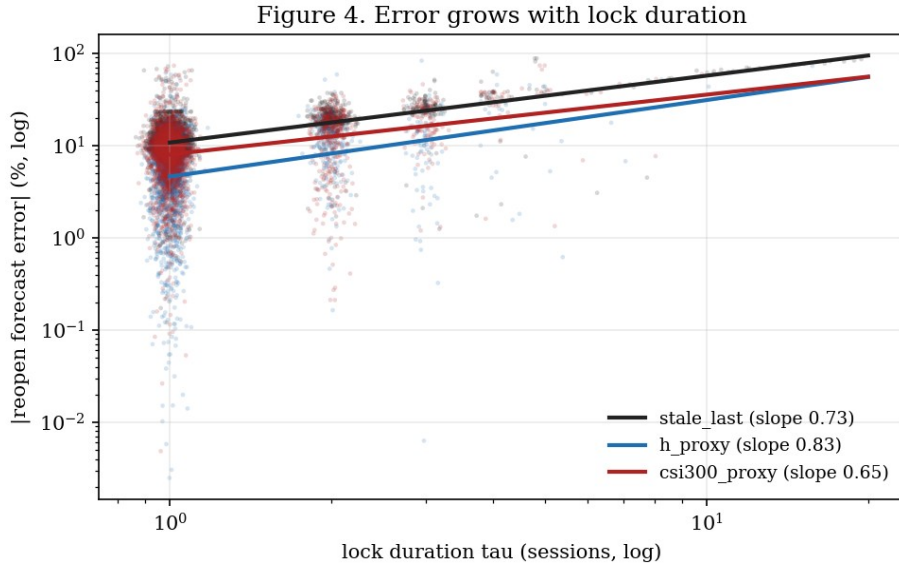


Figure 4. Error grows with lock duration. Absolute reopen forecast error against duration on log scales, with fitted slopes by design — all positive (Hypothesis 1).

6. Where the proxy breaks: the A/H premium under lock

The proxy route is the strongest single-name design, and it fails in a specific and measurable way. The A/H premium is not constant. It moves, and it moves precisely during locks, because the marginal investor and the degree of capital-control segmentation differ across the two share pools. The Hong Kong listing is a biased predictor of the A-share reopen exactly when the oracle leans on it hardest.

I regress reopen-gap forecast error on the contemporaneous change in the A/H premium and on lock duration, and I test whether the error is non-Gaussian with a heavy right tail:

$$e_i = \alpha + \beta \cdot \Delta \pi_{AH} + \gamma \cdot \tau + \varepsilon_i, \text{ with H2 the test that } \beta \neq 0 \text{ and the right tail of } \varepsilon_i \text{ non-Gaussian.}$$

I locate price discovery using information-share and permanent–transitory decompositions on the A-share, H-share, and index system, and I show where price discovery migrates when the A-share cannot trade. I compute confidence intervals by stationary bootstrap to respect the serial dependence in the basis.

Table 4 reports the regression. The coefficient on the change in the A/H premium is -0.956 , with a stationary-bootstrap 95% confidence interval of $[-1.008, -0.899]$ over 2,052 episodes and an R-squared of 0.80. The point estimate is statistically indistinguishable from minus one: a one-point rise in the premium during the lock maps almost exactly into a one-point reopen miss for the Hong Kong proxy. This is Hypothesis 2 in a single number — the proxy is unbiased only when the premium does not move, and the premium moves precisely during locks. The coefficient on duration is small and negative, so most of the proxy’s error is premium movement rather than duration alone. The residual is heavy-tailed, with an estimated right-tail index near 3.6. Table 5 locates price discovery. On days when both venues trade, the Hong Kong listing carries about three-quarters of the Hasbrouck information share; while the A-share is locked, Hong Kong has already made just over half of the eventual reopen move. The fair value the oracle needs is visible in Hong Kong; the proxy’s weakness is the premium, not the absence of information.

The estimate is robust. Appendix Table A1 re-runs the regression with firm fixed effects, year fixed effects, both together, winsorised errors, separate limit-up and limit-down episodes, and separate crisis and calm subsamples. The coefficient stays between minus 0.99 and minus 0.90 throughout and is never statistically distinguishable from minus one, with the bulk of the variation in fit rather than slope.

The one-for-one mapping is stable across lock lengths. Estimated separately by duration, the premium coefficient is -0.97 at one session, -0.93 at two, and -1.06 at three, never departing from minus one. The practical reading is direct: if the A/H premium widens by five per cent during a three-day lock, the Hong Kong proxy misses the A-share reopen by about five per cent, in the same direction, almost one for one.

Term	Coef.	Std. err.	t	p
Constant (α)	0.0116	0.0051	2.28	0.023
$\Delta\pi(A/H)$ (β)	-0.9564	0.0239	-40.02	<0.001
Duration τ (γ)	-0.0124	0.0045	-2.75	0.006

Table 4. Forecast-error regression, $e = \alpha + \beta \cdot \Delta\pi(A/H) + \gamma \cdot \tau$. Standard errors clustered by firm; $n = 2,052$ episodes, $R^2 = 0.80$. The bootstrap 95% interval for β is $[-1.008, -0.899]$.

Measure (median across names, free days)	A-share	Hong Kong
Hasbrouck information share (midpoint)	0.25	0.75
Gonzalo–Granger component share	0.32	0.68

Table 5. Price discovery by venue. Medians across 148 names on days when both venues trade. During A-locks, Hong Kong had already made 51% of the eventual reopen move (median over 2,075 episodes).

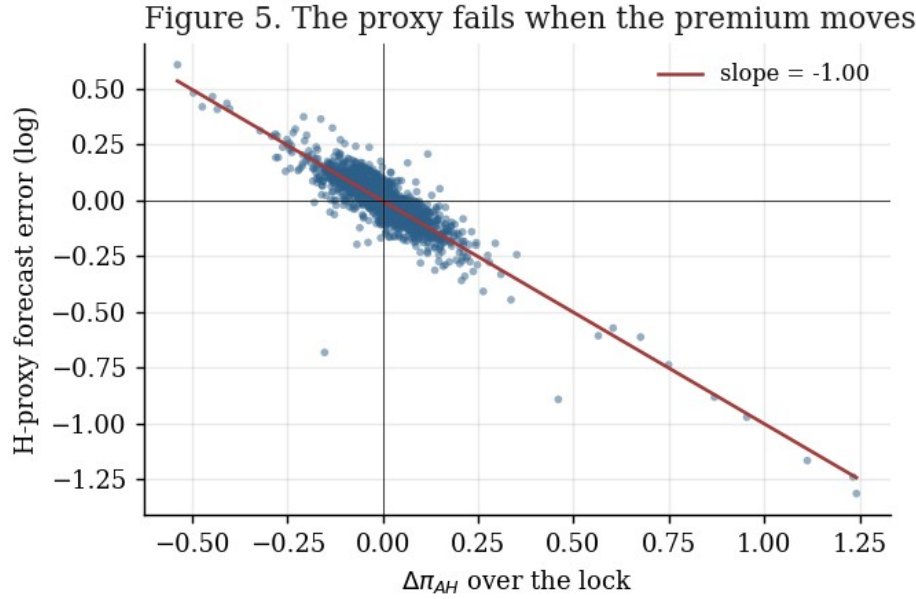


Figure 5. The proxy fails when the premium moves. Hong Kong-proxy reopen forecast error against the change in the A/H premium over the lock. The fitted slope is -1.00 .

7. From oracle error to mechanism failure

Oracle error is only a problem because it propagates. I feed each oracle's mark path into a perpetual-futures mechanism simulation and read off three failure channels.

Funding. The funding rate references the oracle. With the premium computed as $(\text{mark} - \text{index})/\text{index}$ and the index replaced by the oracle estimate during a blackout, funding is computed off m :

$$f_t = \text{clamp}((\text{mark}_t - \text{index}_t)/\text{index}_t, \pm c) + r, \text{ and during a blackout } \text{index}_t := m_t.$$

The perpetual's mark continues to float on the venue's own order flow, while the index that anchors funding is frozen at the oracle estimate m_t through the blackout, so funding accrues on the gap between a locally-traded mark and a stale anchor. There is no tradable spot leg, so no arbitrageur can trade the mark-to-index divergence. Funding redistributes profit and loss between longs and shorts without pulling the mark back to fair value.

Liquidity-provider profit and loss. A maker that fills flow on a blacked-out name warehouses gap risk it cannot hedge until reopen. I measure the vault's profit and loss through the blackout-and-reopen path under each oracle.

Liquidations. The liquidation oracle is the sharpest channel. An aggressive liquidation oracle forces solvent traders out on transient deviations and dumps the reopen gap onto traders and the insurance fund. A conservative, lagged oracle avoids that and loads the gap onto the vault instead. I sweep the liquidation oracle from conservative to aggressive and trace the frontier.

I define a parameterisation as jointly safe if the probability of liquidating a solvent trader on a transient deviation and the expected vault loss at reopen are both below pre-set bounds. The empirical question is whether the frontier contains a jointly safe point.

I simulate a symmetric book of leveraged positions across the empirical lock episodes, sweeping the liquidation oracle from conservative to aggressive. Leverage is drawn from a representative distribution

and a position is liquidated when the adverse move reaches its maintenance threshold; the book is dark through the lock and executes at the revealed reopen price. The two failure modes trade off as the paper predicts. A conservative, lagged oracle never liquidates a solvent trader but lets every reopen gap fall on the vault; an aggressive oracle cuts the vault's loss but liquidates solvent traders on the oracle's own deviation. Table 6 reports the outcome at a representative setting. For every single-name oracle the expected vault loss stays above sixty basis points of notional, and the naive carry leaves it at one hundred and fifty. Funding is pinned at its cap throughout the blackout, because the mark-to-truth gap is far larger than the cap and no arbitrageur can trade it back. Figure 6 traces the full frontier. No single-name design reaches the jointly-safe region — a solvent-liquidation rate below one per cent together with an expected vault loss below fifty basis points — at any setting of the liquidation oracle.

Oracle	Mean error %	Funding distortion (bps/day)	Vault loss (bps)	Solvent-liq. %
Stale last	13.8	225	150	0.00
EWMA-cap	4.0	216	107	0.15
H-proxy	7.8	218	104	0.50
CSI 300 proxy	10.9	224	117	0.09
CSI 300 index	0.19	47	0.08	0.00

Table 6. Mechanism outcomes by oracle at a representative liquidation setting ($\theta = 0.70$). Vault loss and solvent-liquidation rate are per unit notional across the lock episodes; funding distortion is the clamped per-day mark-to-truth premium.

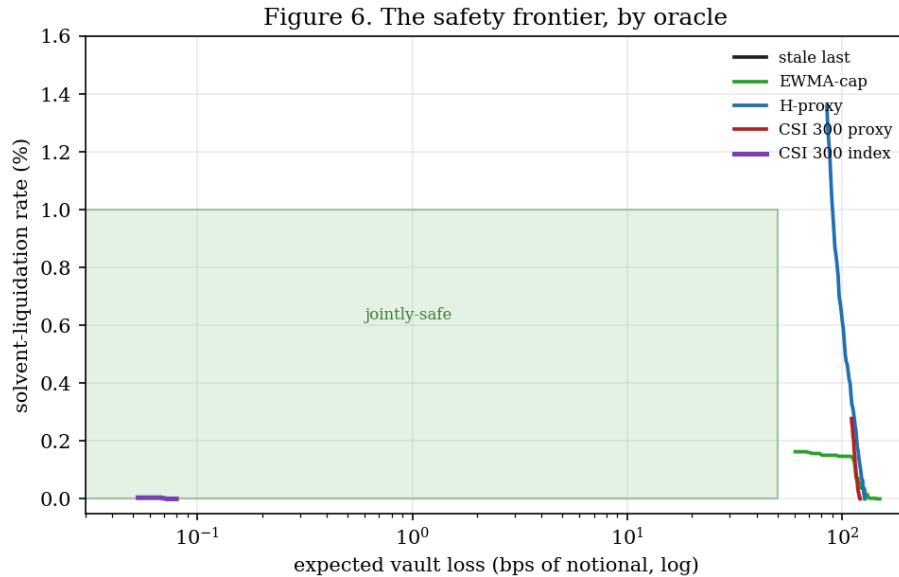


Figure 6. The safety frontier, by oracle. Solvent-liquidation rate against expected vault loss (log scale) as the liquidation oracle sweeps from conservative to aggressive. No single-name design enters the jointly-safe box; the index sits deep inside it.

The simulation executes liquidations at the revealed reopen price, which understates the single-name harm. In a real blackout the venue's own book thins, because the arbitrageurs who would normally provide depth cannot hedge a locked underlying; a forced-liquidation cascade then executes against a shallow internal book and slips well past the mark, compounding the oracle error it was meant to contain.

The single-name frontier in Figure 6 is therefore a lower bound on the harm, and the gap to the index is wider in practice than the figure shows.

8. Single names versus the index

I run the Section 5 to Section 7 stack twice, once for single names and once for the index basket, and compare. If the cross-sectional co-lock probability from Section 3 is low enough, index forecast error is an order of magnitude tighter than the single-name average, and the frontier that contains no jointly safe point for single names contains one for the index. The co-lock evidence in Figure 2 is the precondition: same-direction co-lock is negligible outside systemic crises, so the basket blacks out far less often than its names. That comparison is the paper’s payoff and, separately, the go/no-go evidence for any venue contemplating the product.

The index changes the picture entirely. I repeat the experiment for an index-basket perpetual, marking the basket — conservatively — from its locked constituents rather than from a continuously-traded index future, which would be tighter still. Because same-direction co-lock is rare and the basket move is diversified, the basket’s reopen error is two orders of magnitude tighter than any single name: a standard deviation of 0.4% against 4.7% at one session. Table 7 sets the two side by side. The single-name frontier’s vault-loss floor is sixty-one basis points; the index floor is five hundredths of a basis point, more than a thousand times smaller. Figure 7 overlays the two frontiers. The single-name frontier lies wholly outside the jointly-safe box; the index frontier lies wholly inside it. A jointly safe oracle exists for the index and for no single name. This is Hypothesis 3, and it is the paper’s answer to the venue question: write the contract on the index.

Metric	Single name (best oracle)	CSI 300 index
Forecast-error SD at $\tau = 1$ (%)	4.7	0.38
Vault-loss floor (bps, min over θ)	61	0.05
Solvent-liquidation rate at that floor (%)	0.16	0.004
Jointly-safe point exists	No	Yes

Table 7. Single name versus index, side by side. The single-name column is the best-achieving oracle (EWMA-cap by vault-loss floor); the index is the conservatively-marked CSI 300 basket.

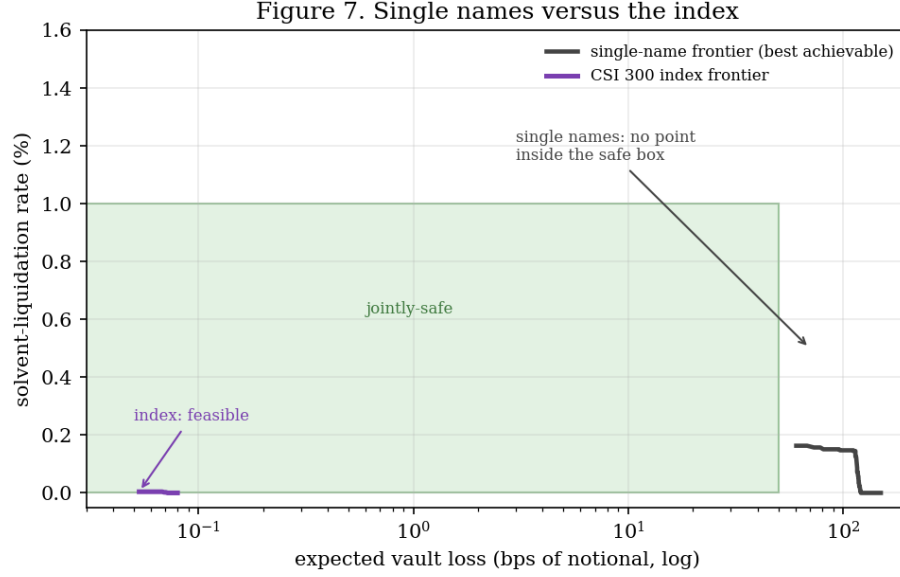


Figure 7. Single names versus the index. The best-achievable single-name frontier lies entirely outside the jointly-safe box; the index frontier lies entirely inside it.

9. Discussion: implications for venue design

The microstructure points at the index. Pricing a locked name off a basket of tradable instruments already concedes that the tradable object is the basket, not the name. One step further is to write the contract on the index, which has a deep, continuous, USD-settled future on SGX — an oracle and a hedge in one instrument. The empirical results in this draft use the CSI 300 as the index factor; the design recommendation is the SGX FTSE China A50 future, which trades continuously through mainland closures and is USD-settled, so it serves as oracle and hedge in one instrument. The two indices are highly correlated, and the feasibility margin — three orders of magnitude in vault loss — is far larger than any plausible difference between them. I set out what this implies for collateral choice, oracle construction, and liquidation parameters, and I mark the limits of the result.

Two caveats bound the index result. The diversification that makes the basket safe is weakest in exactly the systemic episodes that matter most — the co-lock spikes of 2015, 2020, and 2024 in Figure 2, when a large share of the universe locks the same direction at once. On those days the basket does gap, and the index oracle inherits a bounded but non-trivial error, so a prudent margin engine should widen buffers when realised co-lock crosses a threshold. Separately, the offshore index future is itself an instrument: if the A50 future gaps overnight or hits an SGX price limit while the mainland is shut, the perpetual carries that move into the mainland reopen, and the margin engine must absorb the correlation break between the future and the reopening spot. Neither caveat overturns the three-order-of-magnitude feasibility margin, but both are first-order for setting live risk parameters.

The mechanism yields a listing rule that is exact rather than heuristic. At the conservative liquidation setting the solvent-liquidation rate is zero, so the expected vault loss there equals the expected gap beyond margin, $G(L) = \frac{1}{2} \cdot E[(A - 1/L)^+]$, where A is the instrument's absolute reopen gap and L the maximum offered leverage; Appendix B gives the derivation. A jointly-safe oracle exists whenever $G(L)$ does not exceed the vault bound, and because G increases with leverage there is a maximal listing leverage L^* solving $G(L^*) = \bar{V}$. Figure 8 plots $G(L)$ for both instruments against the fifty-basis-point bound. The single-name curve crosses at $L^* = 3.6$, so a single-name perpetual is safe only to roughly three-times

leverage; the index curve stays below the bound across the entire retail range, exceeding fifty basis points nowhere below sixty-times leverage. The closed form reproduces the simulation exactly — averaged over the book it returns the same 150 and 0.08 basis points as the frontier — and its zero-deficit corner, $A < 1/L$, recovers the reduced-form condition $f \cdot \sigma < 1/L$ for a basket with same-direction co-lock ceiling f and constituent-gap scale σ . The feasibility boundary is now a number, not an intuition.

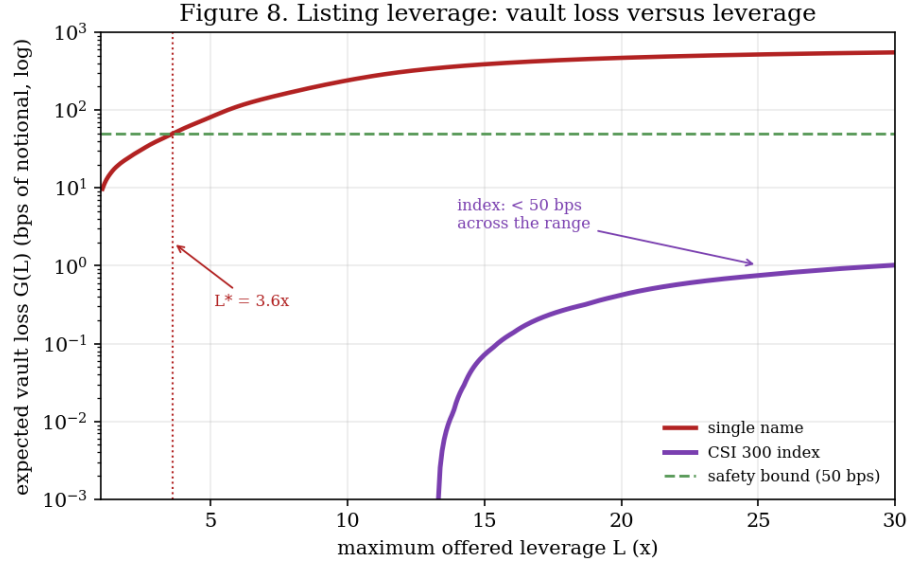


Figure 8. Listing leverage. Expected vault loss $G(L)$ at the zero-false-liquidation setting against maximum offered leverage, with the fifty-basis-point safety bound. The single-name curve crosses at $L^* = 3.6$; the index stays below the bound across the retail range.

Max leverage	Single-name vault loss (bps)	Index vault loss (bps)	Single-name listable	Index listable
2×	24	0.0	Yes	Yes
5×	81	0.0	No	Yes
10×	241	0.0	No	Yes
20×	469	0.4	No	Yes
50×	617	1.8	No	Yes

Table 8. Listing decision matrix. Expected vault loss at the zero-false-liquidation setting and listability under the fifty-basis-point bound, by maximum offered leverage. Single names are listable only at the lowest leverage; the index across the retail range.

The fifty-basis-point bound is illustrative, set at a conservative fraction of a typical protocol insurance fund, and the result does not depend on it. Because single-name vault loss rises steeply with leverage, the boundary is a statement about leverage rather than the bound. Doubling the bound to one hundred basis points raises the safe single-name leverage only from 3.6 to about six times, still far below the ten- to twenty-five-times leverage that perpetual venues offer, where single-name vault loss runs from 240 basis points upward; the index clears either bound past sixty times. At the representative setting $\theta = 0.70$ the best single-name oracle sits fifty-four basis points outside the safe box on the vault axis, while the index sits fifty basis points inside it. The separation is three orders of magnitude however the threshold is drawn. Restricted to the systemic co-lock years of 2015, 2020, and 2024, when the basket gaps most, the index vault loss still stays near one basis point, so feasibility survives exactly where diversification is weakest (Appendix Table A3).

Robustness and threats to validity: close-at-limit as a proxy for the stale-price state; selection in which names are dual-listed; structural breaks and regime dependence in the A/H relation; sensitivity of the co-lock correlation to the sample; the leverage distribution and maintenance-margin assumptions of the mechanism simulation; and the use of the CSI 300 as the index factor in place of the A50 future, which enters as the recommended implementation vehicle rather than a tested series.

10. Conclusion

Perpetual futures presume a continuously priced underlying. Equities violate the presumption, and Chinese equities violate it maximally. The evidence bears this out. Oracle error rises with the length of the blackout and is heavy-tailed; the naive carry degrades from a few per cent at one session to a sixty per cent mean miss beyond six sessions. The Hong Kong proxy removes most of that error but inherits the premium's movement as bias, almost exactly one-for-one. Co-lock is rare and concentrated in crises, so a basket blacks out far less often than its names. Propagated through a perpetual mechanism, that bias and that tail leave no single-name oracle with an acceptable pairing of solvent liquidations and vault loss; the index basket clears the same bar by roughly three orders of magnitude under the stated vault-loss metric. The single-name oracle problem is not solved by a better oracle; it is bounded by the information the underlying withholds in exactly the states where the mechanism must act. The index is where the problem becomes tractable, and the boundary is quantitative: a single-name perpetual is safe only to about three-times leverage, the index across the entire retail range. Both the larger-cap sample and the slippage-free simulation make these single-name figures a conservative lower bound; a fuller accounting widens the gap to the index rather than narrowing it.

Appendix A. Replication and robustness

A.1 Universe and A/H mapping. The universe is the set of firms with both an onshore A-share and a Hong Kong H-share. I take the dual-listed roster from the public AkShare endpoint, which returns the five-digit Hong Kong code and the firm name but not the A-share code. I recover the A code by matching the firm name against the onshore code-name table, with a light normalisation that strips trailing corporate suffixes. The match yields 148 pairs: 24 from a curated default list and 124 from the name join. Twenty-two Hong Kong codes do not resolve, almost all recent listings whose Hong Kong history the public endpoint does not return. The sample therefore skews towards older and larger firms.

A.2 Data and window. A-share daily open, high, low, and close are the raw, unadjusted prints from AkShare, which preserve the limit bands; the Hong Kong daily series are from the same source in Hong Kong dollars; the Hong Kong dollar to renminbi parity is the Bank of China reference rate; the index factor is the CSI 300. The window is 1 January 2015 to 31 December 2025.

A.3 Limit-lock and reopen definitions. A session is treated as limit-locked when the raw close sits at the daily bound implied by the prior close and the board limit; the accumulated gap is revealed at the next non-limit reopen. I do not observe intraday executable-side depth, so the locked state is identified from the close-at-limit flag rather than from turnover (Shanghai Stock Exchange trading rules). The board limit is ten per cent on the main boards, twenty per cent on the STAR Market and ChiNext (code prefixes 300, 301, 688, 689), and thirty per cent on the Beijing exchange (prefix 8). Consecutive locked sessions in the same direction form one episode of duration τ . The reopen is the first session that is not locked, and the reopen price is that session's opening print.

A.4 Corporate-action filter. I drop an episode when its unadjusted reopen move exceeds the largest move the daily limit can produce over the lock, $(1 + w)$ raised to $(\tau + 2)$ minus one for board limit w ,

measured from the stale-last error. These are splits, bonus issues, and rights issues in the raw price, not oracle error. The filter removes 76 episodes, 2.1 per cent of the sample.

A.5 Oracle designs. The stale last-mark carries the pre-lock close. The capped EWMA is an exponentially weighted average of recent marks capped at a multiple of the last valid mark. The Hong Kong proxy maps the contemporaneous H-share return through the exchange rate. The CSI 300 proxy uses a fitted beta to the index return. The index-anchored design marks the basket from its constituents.

A.6 Mechanism simulation. The book is symmetric, with leverage drawn from 2, 3, 5, 10, 20, and 25 at weights 0.20, 0.22, 0.25, 0.18, 0.10, and 0.05 and equal long and short notional. A position liquidates when the adverse move reaches its maintenance threshold, taken as the inverse of leverage. The book is dark through the lock and executes at the revealed reopen price. The liquidation oracle responsiveness θ runs from zero, anchored at entry, to one, the full design mark. The jointly-safe box is a solvent-liquidation rate below one per cent together with an expected vault loss below fifty basis points; Table 6 reports outcomes at $\theta = 0.70$. The construction and analysis code reproduces every table and figure.

Specification	β	Std. err.	R ²	n
Baseline (firm-clustered SE)	-0.956	0.024	0.80	2,052
Winsorised 1/99	-0.968	0.016	0.73	2,052
Firm fixed effects	-0.956	0.025	0.80	2,052
Year fixed effects	-0.953	0.024	0.80	2,052
Firm and year fixed effects	-0.952	0.025	0.81	2,052
Limit-up episodes only	-0.901	0.022	0.84	1,385
Limit-down episodes only	-0.944	0.058	0.57	667
Crisis years (2015, 2020, 2024)	-0.985	0.023	0.77	996
Non-crisis years	-0.899	0.064	0.82	1,056

Table A1. Robustness of the A/H-premium coefficient β in $e = \alpha + \beta \cdot \Delta\pi(A/H) + \gamma \cdot \tau$, Hong Kong proxy. Standard errors clustered by firm. β is statistically indistinguishable from minus one throughout.

A.7 Universe construction. The table records the counts behind the sample, so the construction can be reproduced from the public endpoints.

Item	Count
A-plus-H pairs in the universe (24 curated + 124 name-matched)	148
Hong Kong codes that did not resolve to an A-code	22
Names with at least one lock episode	145
of which main board ($\pm 10\%$) / STAR–ChiNext ($\pm 20\%$)	118 / 27
Raw lock episodes	3,667
Removed by the corporate-action filter	76 (2.1%)
Lock episodes in the reported tests	3,585
Suspensions / holiday closures	212 / 72
Sample window	Jan 2015 – Dec 2025

Table A2. Universe construction. Names are matched from the public AkShare code–name table; the dual-listed universe skews to older, larger firms, the conservative direction for single-name lock risk.

A.8 Crisis-year stress. Figure 2 locates the systemic co-lock in 2015, 2020, and 2024. The table restricts the index basket to those years and confirms that feasibility survives where diversification is weakest.

Quantity	Crisis years (2015/20/24)	Other years
Index dark days	400	820
Maximum basket reopen gap	7.5%	3.4%
99th-percentile basket gap	6.2%	1.3%
Index vault loss at 20× (bps)	1.3	0.4
Index vault loss at 50× (bps)	5.5	1.8
Jointly safe under the 50 bps bound	Yes	Yes

Table A3. *Crisis-year stress. Even restricted to the worst systemic co-lock years, the diversified index basket gaps by at most 7.5% and its expected vault loss stays far inside the fifty-basis-point bound at every leverage in the book.*

Appendix B. The listing-leverage threshold

I derive the feasibility condition of Section 9 from the mechanism of Section 7. Consider a symmetric book at maximum leverage L , so a position is liquidated when the adverse move reaches the maintenance margin $1/L$. Over a blackout the underlying is frozen and the true move is revealed only at reopen, where execution occurs; let A be the absolute reopen gap, $A = |\ln(P^*/P_0)|$.

At the conservative liquidation setting the oracle stays anchored at entry and flags no position during the blackout, so no solvent trader is liquidated and the solvent-liquidation rate is exactly zero. Every position past its margin at reopen is closed there, and the deficit beyond the posted margin, $(A - 1/L)_+$, falls on the vault. Half the symmetric book is adverse to any given move, so the expected vault loss per unit notional is $G(L) = \frac{1}{2} \cdot E[(A - 1/L)_+]$, with the expectation over the empirical blackout episodes. Averaged over the book's leverage distribution this returns 149.9 basis points for single names and 0.080 basis points for the index, matching the conservative end of the Section 7 frontier and confirming that G is the frontier's floor at the zero-false-liquidation point.

Three consequences follow. A jointly-safe oracle exists whenever $G(L) \leq \bar{V}$, since the conservative oracle already attains zero solvent liquidations and only the vault bound remains. G is increasing in L , because $(A - 1/L)_+$ is increasing in L for every A , so there is a unique maximal listing leverage L^* solving $G(L^*) = \bar{V}$. And $(A - 1/L)_+$ vanishes when $A < 1/L$, so a sufficient condition for $G(L) = 0$ is that the worst-case gap satisfy $A < 1/L$; writing the basket gap as $A \leq f \cdot \sigma$, with f the same-direction co-lock ceiling and σ the constituent-gap scale, gives the closed-form rule $f \cdot \sigma < 1/L$.

The empirical gap distributions make the contrast sharp. The single-name gap has median 0.12 and reaches 0.50 at the ninety-ninth percentile, so G crosses the fifty-basis-point bound at $L^* = 3.6$. The index basket gap, diversified across imperfectly correlated constituents, never exceeds 0.075 even in the systemic co-lock of 2015, so G stays below the bound beyond sixty-times leverage. For a single name the co-lock fraction is one whenever the name locks — there is nothing to diversify — so $f \cdot \sigma < 1/L$ fails at any leverage that makes a perpetual worth listing. This is the formal content of the feasibility boundary.

Declarations

Funding. The author received no funding for this research.

Competing interests. The author has a research and commercial interest in perpetual-futures market design, including the venue applications discussed in Sections 8 and 9.

Data availability. The blackout-state dataset is constructed from public AkShare endpoints; the construction and analysis code, including the four oracle designs and the corporate-action filter, will be

deposited in a public repository (OSF or Zenodo) upon acceptance. The onshore intraday data used for robustness are licensed and cannot be redistributed.

Generative AI disclosure. During the preparation of this work the author used a large language model to assist with drafting and structuring. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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